What Predicts Large Funding Rounds for Indian Start-ups?

A Classification and Sector-Clustering Case Study with Classical Machine-Learning Models

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Abstract. Start-ups create jobs and innovation, yet funding is concentrated by city, sector and investor type. This case study asks which observable characteristics of a funding round (city, industry vertical, investment type and year) predict whether an Indian start-up round is *large* (above the median), and whether sectors fall into groups with distinct funding behaviour. We use the public Kaggle file "Indian Startup Funding" (3,044 records, 2015–January 2020), repair its inconsistent dates, amounts and category labels, and keep the 2,073 rounds with a disclosed amount. Rounds from 2015–2017 ($n = 1{,}696$) train the models and 2018–January 2020 ($n = 377$) test them; "large" means above the training median (USD 1.2 million). Logistic regression, a decision tree, Bernoulli naive Bayes and k-NN, tuned with forward-chaining cross-validation, reach a test ROC-AUC of 0.80–0.82 (overlapping confidence intervals) and a Matthews correlation of 0.35–0.58. Ablation shows that investment type carries essentially all of the signal (AUC 0.807 with type alone vs. 0.817 with all four features); within a given investment type, city, sector and year do not rank rounds better than chance (AUC 0.54 for private-equity and 0.45 for seed/angel rounds). The test period is strongly shifted (72.9% large vs. 49.8% in training), so that a constant "always large" predictor reaches F1 = 0.844, above three of the four models, which makes F1 alone misleading here. k-means on 13 sector profiles gives $k = 2$ with a silhouette of only 0.244,
i.e. weak structure. Three hubs (Bengaluru, Delhi-NCR, Mumbai) received 84.6% of disclosed funding, and the top 1% of rounds 44.2%. Simple models are therefore adequate to recover the stage effect, but not to explain geographic or sectoral concentration.

Keywords: start-up funding; India; classification; k-means; logistic regression; decision tree; time-ordered evaluation; SDG 8; SDG 9

1 Introduction

Start-ups are a source of employment and innovation, which is why the United Nations 2030 Agenda links entrepreneurship and small-firm access to finance to SDG 8 (Decent Work and Economic Growth, target 8.3) and SDG 9 (Industry, Innovation and Infrastruc ture, target 9.3) [1]. Funding, however, is not spread evenly: a hand ful of cities, sectors and investor types attract most of the capital. For policy makers who want to widen access, and for founders who want to understand how rounds are sized, a basic question is how much of a round's size can be anticipated from a few public descrip tors.

This paper treats the Indian start-up ecosystem as a single case and asks three questions. **RQ1:** Can city, industry vertical, invest ment type and year classify a funding round as large or small? **RQ2:** Which of these features actually carry the signal? **RQ3:** Do sectors group into clusters with distinct funding behaviour? We use only classical algorithms from an undergraduate syllabus (logistic regres sion, decision tree, naive Bayes, k-NN and k-means) and evaluate them chronologically, so that the models are tested on later years than they were trained on. The study is a *case study* rather than a general method paper: one country, one public dataset, and an em phasis on describing and interpreting what the data do and do not support.

Our contributions are: (i) a documented cleaning of a very messy public dataset (dates, Indian-style number formatting, hundreds of spelling variants of cities, investment types and verticals); (ii) a leakage-free chronological comparison of four classifiers against trivial baselines, reported with bootstrap confidence intervals and metrics that remain meaningful under class-balance shift; (iii) a feature-group ablation and within-type analysis that identify what predicts round size; and (iv) a k-means analysis of sector funding behaviour with an honest assessment of cluster quality. Section 2 reviews related work, Section 3 the data, Sections 4–5 the methods and set-up, Section 6 the results and Section 7 the discussion.

2 Related Work

Research on predicting start-up outcomes has expanded with large platforms such as Crunchbase, whose research use is documented by Dalle et al. [2]. Mashhadi et al. [3] build a leakage-safe firm-quarter panel from Crunchbase for 2010–2023, preprocess on a de-velopment window only, and compare logistic regression with tree ensembles; for the probability of raising another round within 12 months their best model reaches an AUROC of 0.817, and they find that financing recency and firm age dominate. Their task (will a firm raise again?) differs from ours (how large is a given round?), and their features (round history, investor counts, age) are not available in our file. Survey evidence from venture capitalists suggests that investors weigh founding-team quality heavily [4], which is also absent from public round-level tables.

For evaluation on time-ordered data we follow Bergmeir and Benítez [5] and avoid random splits, using forward-chaining cross-validation inside the training years and a strictly later test period. Because class balance shifts over time, we report the Matthews correlation coefficient alongside F1, following Chicco and Jurman [6]. Clusters are produced with Lloyd's k-means algorithm [7] and judged by the silhouette coefficient [8, 9]. All models are imple-mented with scikit-learn [11]; algorithm descriptions follow [12].

Table 1: Columns of startup_funding.csv, raw missing share and their use in this study.

Column	Content	Missing %	Use
Sr No	row index	0.0	dropped
Date	dd/mm/yyyy text	0.0	year, split
Startup Name	text	0.0	not used
Industry Vertical	free text, ≈800 variants	5.6	→ sector (15 classes)
SubVertical	free text	30.7	not used
City Location	free text, 107 variants	5.9	→ city feature
Investors Name	free text	0.8	not used
Investment Type	free text, 55 variants	0.1	→ type (10 classes)
Amount in USD	text with commas	31.5	target
Remarks	text	86.2	not used

3 Dataset and Preprocessing

3.1 Source and description

We used *startup_funding.csv* from the Kaggle dataset "Indian Startup Funding" [13], released under CC0, which lists funding an-nouncements reported for Indian start-ups. The file has 3,044 rows and 10 columns spanning 2 January 2015 to 13 January 2020 (Ta-ble 1). Only 7 rows fall in 2020. We assume, as is usual for such files, that the entries are compiled from published reports rather than from an official registry; we return to the consequences in Section 7.

3.2 Cleaning and encoding

Text cleaning. Several fields contain literal escape sequences (for example non-breaking-space artefacts) and stray whitespace,

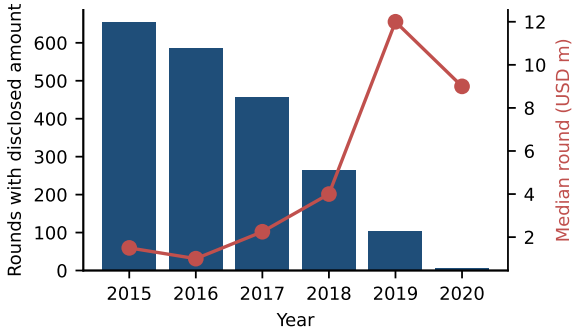


Figure 1: Rounds with a disclosed amount per year (bars) and the median round size (line).

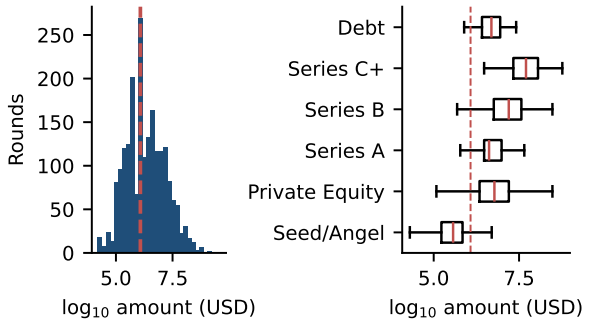


Figure 2: Left: distribution of $\log_{10}$ round size (dashed: training median). Right: round size by investment type.

which we removed before any matching. **Dates.** Eight date strings were malformed (for example a missing separator or a two-digit year fragment); a pattern-based parser repaired all of them. **Amounts.** Amounts are stored as text, some with Indian digit grouping ("20,00,00,000"), a trailing "+" or placeholders ("undisclosed", "unknown", "N/A"). After removing separators, 971 rows (31.9%) had no usable amount and were excluded from modelling; no duplicates were found on date, start-up, investors and amount. This leaves **2,073 rounds**. **Categories.** Spelling variants were merged by rules: Bangalore/Bengaluru, Gurgaon/Gurugram and New Delhi/Delhi are unified (107 raw city strings → 78 canonical names over the full file); 55 raw investment-type strings were mapped to 10 types (Seed/Angel, Private Equity, Pre-Series A, Series A, Series B, Series C+, Debt, Crowdfunding, Other, Unknown); and about 800 vertical strings were mapped by keyword rules to 13 named sectors plus "Other" and "Unknown". Consumer Internet and Technology are broad labels already present in the source and were kept. The rules are listed in the accompanying script. In the model, category levels with fewer than 10 training rows are pooled into one "infrequent" level, and unseen levels at test time fall into that level.

3.3 Descriptive picture

Disclosed round sizes are extremely right-skewed (Fig. 2): the median is USD 1.75 million, the mean 18.4 million, the interquartile range 0.49–8.0 million and the maximum 3.9 billion. The number of reported rounds with an amount falls from 654 in 2015 to 105 in 2019 while the median round grows from USD 1.5 million (2015) to 4.0 million (2018) and 12.0 million (2019) (Fig. 1); the decline in counts probably reflects the file's coverage rather than the ecosystem, so we do not interpret it. Investment type is dominated by two labels: Private Equity (1,082 rounds) and Seed/Angel (865) make up 93.9% of the modelling rows, while the series labels (A, B, C+) appear only from 2018.

Funding is concentrated. Bengaluru, Delhi-NCR (Delhi, Gurugram, Noida and neighbours) and Mumbai account for 75.8% of rounds and 84.6% of disclosed dollars (Bengaluru alone 48.6%); the largest 1% of rounds (21 rounds) hold 44.2% of all disclosed funding, and Private-Equity-labelled rounds hold 71.8% against 2.8% for Seed/Angel. Dollar shares are therefore driven by a few very large rounds. Fig. 3 shows the share of large rounds by city and sector, which are descriptive only because investment type is confounded with both.

3.4 Target and split

The label is $c_i = \mathbf{1}[\text{amount}_i > m]$, with m the median round size in the training years (USD 1.2 million), so that no information from the test period enters the threshold. Rounds are split by time: training = 2015–2017 (1,696 rounds, 49.8% large), test = 2018 to 13 January 2020 (377 rounds, 72.9% large). The jump in the large share is itself a finding: round sizes grew, so a threshold fixed on earlier years labels most later rounds as large. A second shift concerns the type labels: 114 test rounds (30.2%) carry types (Series A/B/C+, Pre-Series A, Other, Unknown, or Debt, which has only 10 training rounds) that are absent or very rare in training.

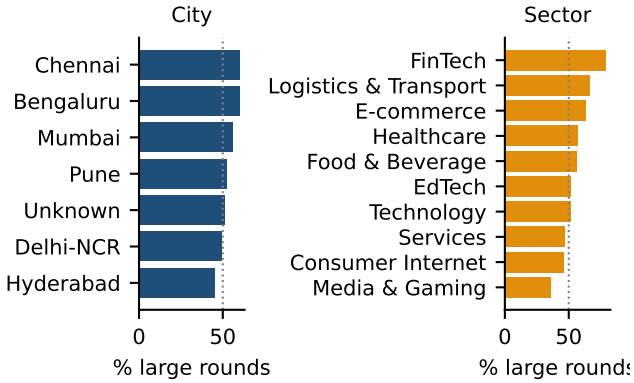


Figure 3: Share of rounds above the training median by city (seven most frequent) and by sector with at least 25 rounds, all years pooled.

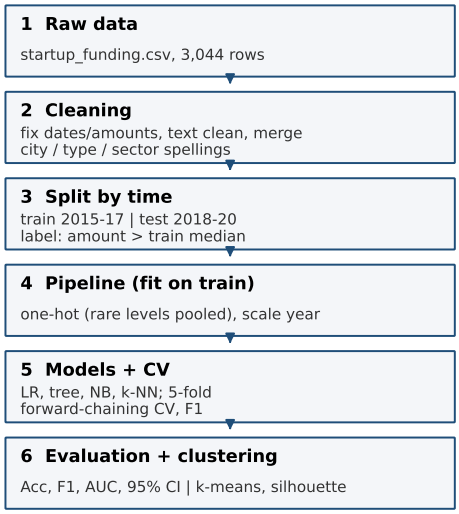


Figure 4: Methodology pipeline used in this study.

4 Methodology

Fig. 4 summarises the workflow. Preprocessing and model fitting are wrapped in a scikit-learn `Pipeline`, so every step is fitted on training data only and re-fitted inside each cross-validation fold. The inputs are city, investment type and sector (one-hot) and year (standardised; binned in three levels for naive Bayes, which needs non-negative features).

Table 2: Hyper-parameter grids and values selected by five-fold forward-chaining CV (criterion: F1).

Model	Grid searched	Selected
Logistic regression	C 0.01, 0.1, 1, 10; class weight none, balanced	C 0.01; class weight none
Decision tree	depth 2, 3, 4, 6, 8; min leaf 5, 20; class weight none, balanced	class weight none; depth 2; min leaf 5
Naive Bayes (Bernoulli)	alpha 0.1, 0.5, 1.0, 2.0	alpha 0.1
k-NN	k 5, 15, 31, 61; weights uniform, distance	k 15; weights uniform

4.1 Baselines

We compare against three trivial predictors: *always small*, *always large*, and a *type-lookup rule* that predicts the training majority class of the round's investment type (unseen types receive the global training rate of 49.8%, which predicts "small"). The first two are the reference points that any model must beat under class-balance shift.

4.2 Classifiers

Logistic regression models $P(c=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, $\sigma(z) = 1/(1+e^{-z})$, and minimises the L2-regularised cross-entropy with inverse strength C. The **decision tree** chooses splits that minimise Gini impurity $G = 1 - \sum_k p_k^2$. **Bernoulli naive Bayes** assumes conditionally independent binary features, $P(\mathbf{x} \mid c) = \prod_j p_{jc}^{x_j}(1-p_{jc})^{1-x_j}$, with Laplace smoothing α; we use it instead of Gaussian naive Bayes because the inputs are one-hot indicators. ***k*-NN** takes a (uniform or distance-weighted) majority vote of the k nearest training rounds under Euclidean distance.

4.3 Clustering sectors

Each sector with at least 10 rounds is described by seven standardised features: median and standard deviation of $\log_{10}$ round size, share of large rounds, share of Seed/Angel rounds, shares of rounds in Bengaluru and in Delhi-NCR, and $\log_{10}$ of the round count. k-means minimises $\sum_{j=1}^{k}\sum_{\mathbf{z}\in C_j}\|\mathbf{z}-\mu_j\|^2$ (50 random starts); $k \in \{2,\dots,6\}$ is chosen by the silhouette $s(i) = \dfrac{b(i)-a(i)}{\max\{a(i),b(i)\}}$ [8], where a and b are the mean distances to the own and the nearest other cluster. Clustering is descriptive and uses all years.

4.4 Evaluation metrics

We report accuracy, balanced accuracy, precision, recall, F1, ROC-AUC and the Matthews correlation coefficient

$$\mathrm{MCC} = \frac{TP\cdot TN - FP\cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}. \qquad (1)$$

Because the test set has 72.9% positives, F1 for the "large" class is inflated by the base rate (a constant "large" prediction scores 0.844), so we emphasise AUC, balanced accuracy and MCC [6]. Uncertainty is quantified with 95% percentile bootstrap intervals (1,000 resamples of the test rounds) [10].

5 Experimental Setup

Tuning. Hyper-parameters were chosen by grid search with five-fold forward-chaining cross-validation (`TimeSeriesSplit`) on the training years, optimising F1 (Table 2); the test set was used once. **Environment.** Python 3.14 on Windows with scikit-learn [x.y.z], pandas, NumPy, SciPy and Matplotlib; random seed 42. The script `run_experiments.py` regenerates every number, table and figure in this paper from the CSV file.

6 Results

6.1 Classification of large rounds

Table 3 and Fig. 5 give the test results. All learned models rank rounds similarly well: AUC lies between 0.799 and 0.817 and the intervals overlap widely (about 0.74–0.87), so the data cannot separate them. Differences appear at the 0.5 decision threshold. The

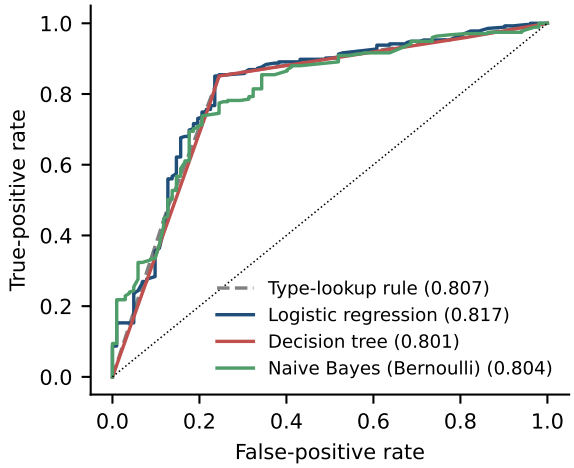


Figure 5: ROC curves of the logistic regression, tree, naive Bayes and the type-lookup rule on the test set (AUC in brackets); the k-NN curve (AUC 0.799) is omitted for clarity.

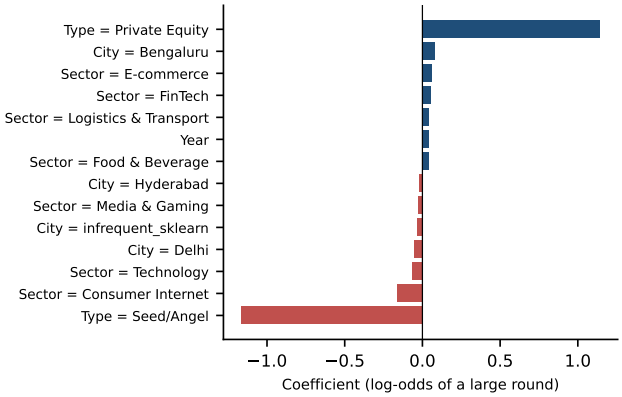


Figure 6: Largest logistic-regression coefficients (log-odds of a large round).

decision tree (depth 2) has the best accuracy (0.825), balanced accuracy (0.803), MCC (0.580) and F1 (0.876); logistic regression and k-NN give MCC of about 0.48; naive Bayes and the type-lookup rule are identical (MCC 0.351) because both reduce to the type indicator and send unseen types to "small". The tuned tree is a single split on Seed/Angel (importance 0.995): non-Seed/Angel rounds are predicted large, which is correct for 89.5% of the 114 test rounds with new or rare types. For logistic regression the confusion matrix was TN = 82, FP = 20, FN = 74, TP = 201 (tree: 77, 25, 41, 234).

Two cautions apply. First, the constant "always large" predictor has F1 = 0.844, which is higher than logistic regression (0.810), k-NN (0.812) and naive Bayes; only the tree beats it on F1, and every learned model beats it on balanced accuracy and MCC (which are 0.5 and 0 for any constant predictor). Second, performance in the training period was much higher than on the test set: forward-chaining CV gave F1 $\approx$ 0.89 and ROC-AUC 0.892 for logistic regression against a test AUC of 0.817, a sign of the temporal shift described in Section 3.4.

6.2 What predicts a large round?

The logistic-regression coefficients (Fig. 6) are dominated by investment type: Private Equity raises the odds of a large round by a factor of 3.13 (coefficient +1.14) and Seed/Angel lowers them to 0.31 (−1.17). The next largest effects are small: Consumer Internet (−0.16, odds ratio 0.85), Bengaluru (+0.08) and E-commerce (+0.06). The strong regularisation selected by cross-validation ($C = 0.01$) shrinks these further, so they should not be read as evidence of city or sector effects.

The ablation (Table 4, Fig. 7) confirms this. Investment type

Table 3: Large-vs-small funding-round classification on the 2018–Jan 2020 test set ($n = 377$; 72.9% large). CV-F1 is the mean over five forward-chaining folds inside 2015–2017. Best learned-model value in bold; 95% bootstrap CIs (1,000 resamples).

Model	Acc.	Bal. acc.	MCC	Prec.	Recall	F1	F1 95% CI	AUC	AUC 95% CI	CV-F1
Always 'small'	0.271	0.500	0.000	0.000	0.000	0.000	[0.000, 0.000]	–	–	–
Always 'large'	0.729	0.500	0.000	0.729	1.000	0.844	[0.813, 0.871]	–	–	–
Type-lookup rule	0.623	0.696	0.351	0.908	0.538	0.676	[0.622, 0.724]	0.807	[0.752, 0.858]	–
Logistic regression	0.751	0.767	0.482	0.910	0.731	0.810	[0.769, 0.845]	**0.817**	[0.758, 0.867]	0.892
Decision tree	**0.825**	**0.803**	**0.580**	0.903	0.851	**0.876**	[0.843, 0.905]	0.801	[0.748, 0.849]	0.892
Naive Bayes (Bernoulli)	0.623	0.696	0.351	0.908	0.538	0.676	[0.622, 0.724]	0.804	[0.748, 0.853]	0.892
k-NN	0.751	0.761	0.473	0.902	0.738	0.812	[0.771, 0.846]	0.799	[0.737, 0.856]	0.896

Table 4: Feature-group ablation with the tuned logistic regression (same hyperparameters, refitted). AUC on the test set and mean forward-chaining CV AUC.

Feature set	Test AUC	CV AUC	Test F1
All four features	0.817	0.892	0.810
Without City	0.812	0.897	0.824
Without Investment type	0.586	0.564	0.847
Without Sector	0.830	0.888	0.629
Without Year	0.814	0.895	0.632
Only City	0.540	0.554	0.612
Only Investment type	0.807	0.893	0.629
Only Sector	0.541	0.577	0.635
Only Year	0.598	0.488	0.844

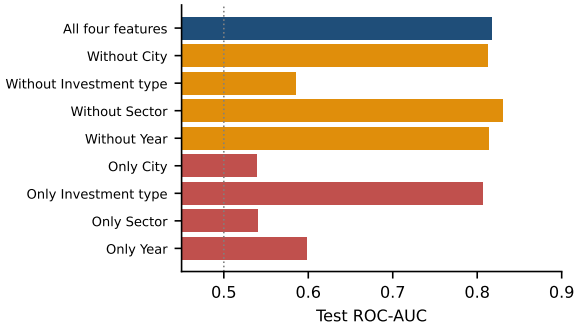


Figure 7: Test ROC-AUC of the tuned logistic regression for different feature sets.

Table 5: Test rounds whose investment type was seen in training (Private Equity or Seed/Angel; $n = 263$)

Model	Acc.	MCC	F1	AUC
Always 'large'	0.658	0.000	0.794	–
Type-lookup rule	0.795	0.590	0.830	0.809
Logistic regression	0.795	0.590	0.830	0.820
Decision tree	0.795	0.590	0.830	0.806
Naive Bayes (Bernoulli)	0.795	0.590	0.830	0.822
k-NN	0.795	0.590	0.830	0.790

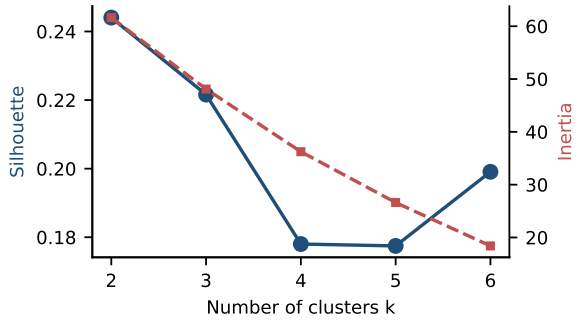


Figure 8: Silhouette (left axis) and inertia (right axis) of k-means for $k = 2, \ldots, 6$.

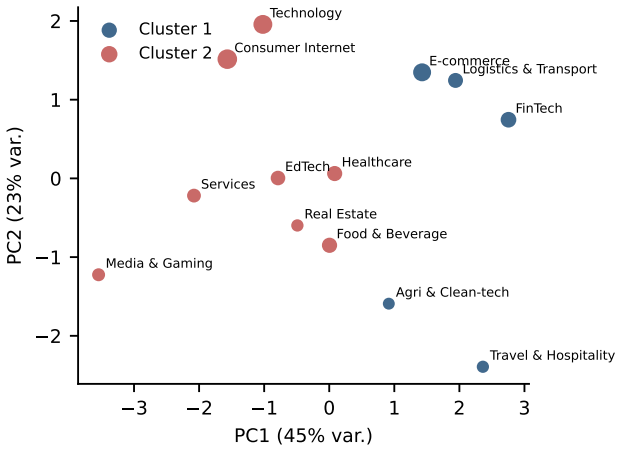


Figure 9: Sectors in the first two principal components of the clustering features (68% of variance); point size grows with the number of rounds.

alone gives a test AUC of 0.807 and a CV AUC of 0.893, essentially the same as all four features (0.817 and 0.892). Removing type drops the AUC to 0.586 (CV 0.564), whereas removing city, sector or year leaves it between 0.812 and 0.830. City, sector and year alone reach only 0.540, 0.541 and 0.598 on the test set, and year alone is below chance in CV (0.488). F1 changes sharply across these variants (0.61–0.85) while AUC stays within 0.81–0.83 whenever type is retained, which suggests that the F1 differences come from where the 0.5 threshold falls for rounds with unseen types rather than from better ranking.

Two further checks (both added after seeing the main results, and therefore exploratory) point the same way. On the 263 test rounds whose type was seen in training, logistic regression, the tree, naive Bayes, k-NN and the lookup rule make identical predictions (accuracy 0.795, MCC 0.590, F1 0.830; Table 5); extra features do not change a single decision. And *within* a type, a logistic model using only city, sector and year ranks held-out rounds no better than chance: AUC 0.539 (95% CI 0.378–0.707, $n = 145$) for Private Equity and 0.448 (0.338–0.550, $n = 118$) for Seed/Angel.

6.3 Sector clusters

Thirteen sectors have at least 10 rounds (1,845 rounds in total). The silhouette is highest at $k = 2$ (0.244) and lower for $k = 3$ to 6 (0.222, 0.178, 0.177, 0.199; Fig. 8). By the usual reading of the silhouette [9], values of 0.25 or below indicate no substantial structure, so the partition is a descriptive summary rather than evidence of natural groups; repeated single-start runs agree only moderately with the final solution (mean adjusted Rand index 0.42).

Table 6 and Fig. 9 show what separates the two groups. Cluster 1 (Agri and Clean-tech, E-commerce, FinTech, Logistics and Transport, Travel and Hospitality; 500 rounds) has larger typical rounds (about USD 5.0 million), a higher large-round share (66.6%) and fewer Seed/Angel rounds (31.8%). Cluster 2 (Consumer Internet, EdTech, Food and Beverage, Healthcare, Media and Gaming, Real Estate, Services, Technology; 1,345 rounds) has smaller typical rounds (about USD 1.5 million) and a larger seed share (47.7%). The split mostly mirrors the stage mix: it is the Seed/Angel share, not the sector name, that moves with round size. Small sectors (16–28 rounds, for example Agri and Clean-tech or Media and Gaming) have noisy profiles.

Table 6: Profile of the two sector clusters ($k = 2$). Large, Seed/Angel, Bengaluru and Delhi-NCR are shares of rounds (means over member sectors); the median amount is the median of the sectors' median round sizes.

Cluster	Sectors	Rounds	Median (USD m)	Large	Seed/Angel	Bengaluru	Delhi-NCR
1	Agri & Clean-tech, E-commerce, FinTech, Logistics & Transport, Travel & Hospitality	500	5.0	66.6%	31.8%	22.8%	37.4%
2	Consumer Internet, EdTech, Food & Beverage, Healthcare, Media & Gaming, Real Estate, Services, Technology	1,345	1.5	50.5%	47.7%	29.6%	32.6%

6.4 Sensitivity to the threshold

Using the pooled median of all rounds (USD 1.75 million) instead of the training median changes the test share of large rounds to 68.7% and the logistic-regression result to AUC 0.849, accuracy 0.623 and F1 0.649. The ranking quality is thus not tied to the exact threshold, while accuracy and F1 at a fixed 0.5 cut-off are.

7 Discussion

What the case shows. In this file, investment type is almost all there is to predict. Seed/Angel rounds are small (5.1% large in training) and Private Equity rounds are large (85.2%), and neither city, sector nor year adds a measurable amount of information on top. Four different algorithms therefore converge on the same decision rule, and the interesting results are about evaluation: under a drifting threshold, the best-looking F1 can come from a constant predictor, and the ranking metric (AUC $\approx$ 0.81) is more stable than the thresholded ones.

Reading the concentration result. Bengaluru, Delhi-NCR and Mumbai attract most of the disclosed dollars, but this does not mean that a city label predicts a bigger round: after accounting for type, the city, sector and year features do not rank rounds better than chance. The geographic concentration is mainly one of deal *counts* and of a few very large rounds (top 1% of rounds: 44.2% of dollars). For SDG 8 and SDG 9 this suggests that widening access to capital is a question of how many early-stage rounds happen outside the three hubs, not of round size once a round occurs; our data cannot test this directly.

Limitations. (i) The file is compiled from reports, not from an official registry, and 31.9% of rows lack a usable amount; disclosure is unlikely to be random, so the results describe *disclosed* rounds. (ii) The type labels are coarse and may encode the round size by construction (a "Seed" round is small by definition), so a model that uses them is close to circular; excluding type leaves little signal (AUC 0.586). (iii) The test set is small (377 rounds), 30.2% of it has unseen types, and its class balance differs from training. (iv) Only four features were used, as the problem statement prescribes; investor identity, start-up age, team and traction, which the literature considers important [3, 4], are not in the data. (v) Sector classes come from keyword rules and Consumer Internet and Technology are broad; the clustering rests on 13 points and has a silhouette below 0.25. (vi) The slice and within-type analyses were added after the main results and should be seen as exploratory. (vii) The data stop in January 2020 and say nothing about later funding cycles.

Ethics and responsible use. The data are public announcements about companies and investors; no personal information beyond organisation names is analysed. The main risk is misuse: using such a model to screen start-ups could reinforce the concentration it describes, and the models here are not suitable for investment decisions.

8 Conclusion and Future Work

Using 2,073 disclosed funding rounds from an Indian start-up file and a chronological, leakage-free protocol, we found that classical models classify large vs. small rounds with a test AUC of about 0.81 (MCC up to 0.58), that this performance comes from the investment-type label alone, and that city, sector and year add no detectable information; sectors show only weak clustering (silhouette 0.244). Future work should add investor and start-up-level features from richer sources, evaluate on data beyond 2020, model the round amount directly, and use a stage taxonomy that does not itself encode size.

Declarations

Data availability. The dataset is publicly available at `https://www.kaggle.com/datasets/sudalairajkumar/indianstartup-funding` . Code and figures: `run_experiments.py` and the `figs/` folder accompanying this report.

Use of AI tools. The analysis code, figures and first draft of this manuscript were produced with an AI assistant (Claude, Anthropic) from the dataset and the assignment template supplied.

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